

1 / 20 | Ages 9-12 | 20 minutes

Does stirring change dissolving?

What we are practising

Plan a fair comparison and distinguish dissolving speed from amount added.

What you need

Two clear cups / Room-temperature water / Table salt / Two teaspoons / Tray

Use sugar instead of salt, using the same small amount in both cups.

Before you begin

Place the cups on a tray and add the same half-cup amount of water to each.

An adult supervises; use no hot water, avoid tasting and wipe spills promptly.

Try it together

1. Predict whether stirring will change how quickly visible grains disappear.
2. Add one level quarter-teaspoon of salt to each cup at nearly the same time.
3. Stir one cup gently for thirty seconds while leaving the other still.
4. Compare visible grains after one minute and describe any uncertainty.

Talk, draw or show your idea

Which details must match so stirring is the main difference?

Space to explore

2 / 20 | Ages 9-12 | 30 minutes

Keep the ice cool

What we are practising

Compare insulation while acknowledging limits in a small experiment.

What you need

Two similar ice cubes / Two identical shallow bowls / Dry folded cloth / Tray

If ice is unavailable, draw the setup and identify what would need to stay constant.

Before you begin

Place bowls side by side on a tray away from sun, heaters and drafts.

An adult handles ice if uncomfortable to touch; keep meltwater contained and never place ice on skin for a prolonged period.

Try it together

1. Put one similar-sized ice cube in each bowl.
2. Wrap one bowl loosely in the dry cloth and leave the other uncovered.
3. Predict the result, then wait twenty minutes while drawing the setup.
4. Uncover once and compare the remaining ice and meltwater.

Talk, draw or show your idea

If both cubes remain partly frozen, what can you still compare?

Space to explore

3 / 20 | Ages 9-12 | 15 minutes

See a vibration

What we are practising

Describe vibration and compare two free lengths under similar conditions.

What you need

Flexible plastic ruler without cracks / Stable table

Observe a gently plucked paper strip and discuss visible vibration without comparing pitch.

Before you begin

An adult checks the ruler and demonstrates holding most of it flat on the table.

Adult supervision is required; use an undamaged flexible ruler, only tiny bends, and keep faces away from the free end. Stop if it cracks or feels stiff.

Try it together

1. Leave about 8 cm beyond the edge, press the rest down firmly and gently flick the free tip downward.
2. Look at the repeated movement and listen at a comfortable distance.
3. Repeat with about 12 cm free, using a similarly gentle flick.
4. Describe changes in vibration and pitch without needing musical note names.

Talk, draw or show your idea

Which free length seems to vibrate faster?

Space to explore

4 / 20 | Ages 9-12 | 20 minutes

A surface that slows a slide

What we are practising

Investigate how a surface can affect sliding while controlling slope.

What you need

Stiff cardboard / One book / Small block or eraser / Dry cloth / Tray / Paper / Pencil

Use a flat safe board instead of cardboard.

Before you begin

Make a low ramp on a tray using one book; mark a release line near the top.

An adult stabilizes the low ramp; use light objects, contain them on a tray and never test on stairs or with people.

Try it together

1. Place the block at the line and release without pushing.
2. Record whether it slides freely, slowly or not at all.
3. Cover the same ramp with cloth and repeat from the same line.
4. Repeat each setup three times without changing ramp height.

Talk, draw or show your idea

Why would giving one trial a push make the comparison less fair?

Space to explore

5 / 20 | Ages 9-12 | 20 minutes

A model of day and night

What we are practising

Use a model to explain changing illumination and identify its limitations.

What you need

Ball / Removable paper marker / Torch

Use a round fruit only as a model, keeping it clean and uneaten during play.

Before you begin

An adult places a cool torch at one fixed position; add a marker to the ball for an imaginary town.

Never look into the torch or the Sun; keep lights cool and paths clear.

Try it together

1. Hold the ball where one side is lit and the other is shaded.
2. Slowly rotate it in place while watching the town marker.
3. Describe when the town enters light, stays lit and moves into darkness.
4. State one thing the model shows well and one thing it does not represent accurately.

Talk, draw or show your idea

Do you need to move the light around the ball to create this cycle?

Space to explore

6 / 20 | Ages 9-12 | 10 minutes

One result or a reliable pattern?

What we are practising

Evaluate repeatability before making a strong claim.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

A paper helicopter takes 2 seconds to fall once; a second design takes 3 seconds once, from the same height.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. A paper helicopter takes 2 seconds to fall once; a second design takes 3 seconds once, from the same height.
2. Ask: Is one fall per design enough to claim the second always falls more slowly? What should come next?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

A paper helicopter takes 2 seconds to fall once; a second design takes 3 seconds once, from the same height. Is one fall per design enough to claim the second always falls more slowly? What should come next?

Space to explore

7 / 20 | Ages 9-12 | 10 minutes

Equal temperature, different feel

What we are practising

Distinguish temperature from heat-transfer rate.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

A metal block and a wooden block both measure 22°C; a warmer hand touches each briefly in this imagined example.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. A metal block and a wooden block both measure 22°C; a warmer hand touches each briefly in this imagined example.
2. Ask: Why might the metal feel cooler even though both temperatures match?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

A metal block and a wooden block both measure 22°C; a warmer hand touches each briefly in this imagined example. Why might the metal feel cooler even though both temperatures match?

Space to explore

8 / 20 | Ages 9-12 | 10 minutes

Which way does heat move?

What we are practising

Predict the direction of heat transfer.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Imagine a room-temperature spoon placed in warmer water, with no heater involved in the activity.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. Imagine a room-temperature spoon placed in warmer water, with no heater involved in the activity.
2. Ask: Initially, does thermal energy move overall from spoon to water or water to spoon?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Imagine a room-temperature spoon placed in warmer water, with no heater involved in the activity.

Initially, does thermal energy move overall from spoon to water or water to spoon?

Space to explore

9 / 20 | Ages 9-12 | 10 minutes

Water inside a sealed system

What we are practising

Apply conservation of matter to a defined boundary.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

In a sealed container, liquid water evaporates and later condenses on the lid; nothing leaks in or out.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. In a sealed container, liquid water evaporates and later condenses on the lid; nothing leaks in or out.
2. Ask: Does evaporation alone reduce the total mass of the entire sealed container and its contents?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

In a sealed container, liquid water evaporates and later condenses on the lid; nothing leaks in or out.

Does evaporation alone reduce the total mass of the entire sealed container and its contents?

Space to explore

10 / 20 | Ages 9-12 | 10 minutes

Opposite seasons on one Earth

What we are practising

Explain seasons using axial tilt.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

In June, the Northern Hemisphere tilts towards the Sun and the Southern Hemisphere tilts away.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. In June, the Northern Hemisphere tilts towards the Sun and the Southern Hemisphere tilts away.
2. Ask: Why can the hemispheres have opposite seasonal patterns at the same time?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

In June, the Northern Hemisphere tilts towards the Sun and the Southern Hemisphere tilts away. Why can the hemispheres have opposite seasonal patterns at the same time?

Space to explore

11 / 20 | Ages 9-12 | 10 minutes

Why the Moon's shape seems to change

What we are practising

Separate lunar phases from eclipses.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Over a month we see crescents, a half-lit disk and a full Moon; consider ordinary phases, not eclipses.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. Over a month we see crescents, a half-lit disk and a full Moon; consider ordinary phases, not eclipses.
2. Ask: Is the Moon changing its physical shape, or are we seeing different amounts of its sunlit half?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Over a month we see crescents, a half-lit disk and a full Moon; consider ordinary phases, not eclipses. Is the Moon changing its physical shape, or are we seeing different amounts of its sunlit half?

Space to explore

12 / 20 | Ages 9-12 | 10 minutes

Where the rolling energy goes

What we are practising

Trace energy beyond the moving object.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

A toy rolls across a level floor and stops without being caught.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. A toy rolls across a level floor and stops without being caught.
2. Ask: Has its energy been destroyed, or transferred into other forms and surroundings?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

A toy rolls across a level floor and stops without being caught. Has its energy been destroyed, or transferred into other forms and surroundings?

Space to explore

13 / 20 | Ages 9-12 | 10 minutes

Can ordinary filter paper catch salt?

What we are practising

Select a separation method according to particle state.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

A mixture contains water, dissolved table salt and visible sand; imagine pouring it through ordinary filter paper.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. A mixture contains water, dissolved table salt and visible sand; imagine pouring it through ordinary filter paper.
2. Ask: Which part is more likely to stay on the filter: sand grains or dissolved salt?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

A mixture contains water, dissolved table salt and visible sand; imagine pouring it through ordinary filter paper. Which part is more likely to stay on the filter: sand grains or dissolved salt?

Space to explore

14 / 20 | Ages 9-12 | 10 minutes

The measurement that stands out

What we are practising

Investigate an outlier without hiding data.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Four readings of one unchanged pencil are 14.1, 14.0, 14.1 and 41.0 centimetres.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. Four readings of one unchanged pencil are 14.1, 14.0, 14.1 and 41.0 centimetres.
2. Ask: What should you do with 41.0 before calculating and reporting a typical length?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Four readings of one unchanged pencil are 14.1, 14.0, 14.1 and 41.0 centimetres. What should you do with 41.0 before calculating and reporting a typical length?

Space to explore

15 / 20 | Ages 9-12 | 10 minutes

Umbrellas and rainy days

What we are practising

Distinguish association from causation.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

A notebook records more open umbrellas on rainy days than dry days.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. A notebook records more open umbrellas on rainy days than dry days.
2. Ask: Does this show that opening umbrellas causes rain?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

A notebook records more open umbrellas on rainy days than dry days. Does this show that opening umbrellas causes rain?

Space to explore

16 / 20 | Ages 9-12 | 10 minutes

Light or leaning pot?

What we are practising

Design a test that distinguishes competing explanations.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

A seedling leans toward a window, but its pot is also tilted. Someone claims light caused all the leaning.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. A seedling leans toward a window, but its pot is also tilted. Someone claims light caused all the leaning.
2. Ask: What should be controlled before this observation can test a response to light direction?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

A seedling leans toward a window, but its pot is also tilted. Someone claims light caused all the leaning. What should be controlled before this observation can test a response to light direction?

Space to explore

17 / 20 | Ages 9-12 | 10 minutes

Estimating a whole patch

What we are practising

Use sampling while acknowledging uncertainty.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

A 20-square-metre patch has four randomly chosen one-square-metre samples containing 2,4,3 and 3 daisies.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. A 20-square-metre patch has four randomly chosen one-square-metre samples containing 2,4,3 and 3 daisies.
2. Ask: What rough estimate follows from the sample mean, and is it an exact count?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

A 20-square-metre patch has four randomly chosen one-square-metre samples containing 2,4,3 and 3 daisies. What rough estimate follows from the sample mean, and is it an exact count?

Space to explore

18 / 20 | Ages 9-12 | 10 minutes

Rusting versus reshaping

What we are practising

Distinguish a chemical change from reshaping.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Compare a paperclip bent into a new shape with iron forming rust in moist air; use drawings.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. Compare a paperclip bent into a new shape with iron forming rust in moist air; use drawings.
2. Ask: Which example forms new substances through a chemical change?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Compare a paperclip bent into a new shape with iron forming rust in moist air; use drawings. Which example forms new substances through a chemical change?

Space to explore

19 / 20 | Ages 9-12 | 10 minutes

Compressing the air model

What we are practising

Explain gas compression with a particle model.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw air in a sealed syringe with no needle; a plunger reduces the enclosed volume without air escaping. This is a diagram only.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. Draw air in a sealed syringe with no needle; a plunger reduces the enclosed volume without air escaping. This is a diagram only.
2. Ask: In a particle model, does compression mainly make the particles smaller or reduce the spaces between them?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw air in a sealed syringe with no needle; a plunger reduces the enclosed volume without air escaping. This is a diagram only. In a particle model, does compression mainly make the particles smaller or reduce the spaces between them?

Space to explore

20 / 20 | Ages 9-12 | 10 minutes

Equal volumes, different masses

What we are practising

Calculate and compare density with consistent units.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Liquid A has mass 10 grams for 10 millilitres; liquid B has mass 12 grams for 10 millilitres, excluding containers.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. Liquid A has mass 10 grams for 10 millilitres; liquid B has mass 12 grams for 10 millilitres, excluding containers.
2. Ask: Which has greater density, and what measurements support that conclusion?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Liquid A has mass 10 grams for 10 millilitres; liquid B has mass 12 grams for 10 millilitres, excluding containers. Which has greater density, and what measurements support that conclusion?

Space to explore

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

1. Does stirring change dissolving?

A little hint

Use the same cups, water amount, temperature and salt amount.

Answer / what to notice

Stirring usually makes the small amount dissolve faster; record what is visible rather than claiming exact times you did not measure.

Stirring brings fresh water into contact with salt; dissolved salt remains in the water even when grains are no longer visible.

Make it easier

Compare the cups by observation without timing.

Take it further

Repeat with fresh mixtures three times and discuss variation between trials.

Another day, away from the screen

Draw particles before and after dissolving, marking the drawing as a simplified model.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

2. Keep the ice cool

A little hint

Compare how much solid ice remains, not just whether any ice remains.

Answer / what to notice

The covered cube may melt more slowly; results depend on cloth, contact and room conditions, so report the observation honestly.

A cloth can slow heat transfer from warmer surroundings; it does not create cold or stop melting forever.

Make it easier

Observe the two setups with an adult describing the difference.

Take it further

Repeat with fresh equal cubes and a different dry wrapping material.

Another day, away from the screen

Explain why an insulated lunch bag still needs an appropriate cooling source.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

3. See a vibration

A little hint

The longer free section is usually easier to bend and vibrates more slowly.

Answer / what to notice

The shorter free section usually makes a higher-pitched sound; the longer one usually makes a lower pitch.

The vibrating ruler moves surrounding air; changing its free length changes its vibration frequency.

Make it easier

Watch one safe demonstration and sketch the motion.

Take it further

Compare three free lengths and record a qualitative high-to-low order.

Another day, away from the screen

Identify an everyday instrument part that vibrates to make sound.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

4. A surface that slows a slide

A little hint

Start each trial in the same place and let go rather than shove.

Answer / what to notice

The cloth often slows or stops the block, but the exact result depends on both materials and the slope.

Friction opposes sliding; changing the contacting surface can change how strongly it resists motion.

Make it easier

Compare only one slide on each surface.

Take it further

Keep the surface fixed and separately investigate the effect of ramp height.

Another day, away from the screen

Name situations where more friction helps, such as a grippy shoe sole.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

5. A model of day and night

A little hint

Keep the light fixed while the ball rotates around its own axis.

Answer / what to notice

The marker experiences light and darkness as the ball turns; Earth's rotation produces the day-night cycle.

The model represents rotation and illumination, but not the real sizes, distances, sunlight rays or exact day lengths.

Make it easier

Compare only the lit and unlit sides without rotating.

Take it further

Mark a second town on the opposite side and compare their illumination at the same time.

Another day, away from the screen

Draw four positions of the same town during one full turn.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

6. One result or a reliable pattern?

A little hint

A release can vary from trial to trial.

Answer / what to notice

No. Repeat comparable trials with the same height and release method, recording every result.

Repeats reveal variation; a single comparison cannot justify “always”.

Make it easier

Compare three imagined results per design.

Take it further

Propose a summary such as the mean and report the spread as well.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

7. Equal temperature, different feel

A little hint

Feeling depends on how quickly heat leaves the hand.

Answer / what to notice

Metal usually transfers heat away from the warmer hand faster than wood.

Touch senses heat transfer as well as conditions at the skin; it is not a direct thermometer reading.

Make it easier

Draw an arrow from the warmer hand into each material.

Take it further

Predict how metal might feel if both objects were hotter than the hand; do not test with hot objects.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

8. Which way does heat move?

A little hint

Compare the two starting temperatures.

Answer / what to notice

From warmer water to the cooler spoon.

Net heat transfer goes from higher to lower temperature until equilibrium is approached.

Make it easier

Label the water warmer and spoon cooler.

Take it further

Explain why the temperature difference tends to shrink over time.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

9. Water inside a sealed system

A little hint

Include invisible vapour as part of the contents.

Answer / what to notice

No; the total mass remains the same within ordinary measurement limits.

Water changes state and location inside the boundary instead of leaving the system.

Make it easier

List liquid, vapour and droplets as forms of the same water.

Take it further

Explain why an open container can lose measured mass by evaporation.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

10. Opposite seasons on one Earth

A little hint

Think about sunlight angle and daylight duration.

Answer / what to notice

Earth's tilted axis gives the hemispheres different sunlight angles and day lengths.

Distance from the Sun alone cannot explain opposite seasons on the same planet at the same time.

Make it easier

Mark the two hemispheres on a drawn globe.

Take it further

Explain why local weather does not have to match a season's usual pattern every day.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

11. Why the Moon's shape seems to change

A little hint

Keep the Moon spherical in every sketch.

Answer / what to notice

We see different portions of its sunlit half as it orbits Earth.

Changing Sun-Earth-Moon geometry produces phases; Earth's shadow explains lunar eclipses, not ordinary phases.

Make it easier

Draw a half-lit sphere viewed from two directions.

Take it further

Explain why the unseen far side is not permanently dark.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

12. Where the rolling energy goes

A little hint

Friction and sound can transfer energy.

Answer / what to notice

It has been transferred, including small increases in thermal energy and some sound.

The toy's motion decreases, but that does not mean total energy disappears.

Make it easier

Draw arrows from toy to floor and surrounding air.

Take it further

Explain why we might not notice the tiny temperature changes.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

13. Can ordinary filter paper catch salt?

A little hint

Compare grain size with dissolved particles.

Answer / what to notice

Sand grains; dissolved salt generally passes through with the water.

Ordinary filtration separates larger suspended particles, not dissolved ions. The filtered water is not for drinking.

Make it easier

Draw sand caught above and salty water below.

Take it further

Propose evaporation as a way to recover salt on paper, without heating anything.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

14. The measurement that stands out

A little hint

A surprising value deserves checking, not silent deletion.

Answer / what to notice

Recheck the ruler reading and record; it may be a transposition, but do not assume that without checking.

Keeping a note of any correction makes the measurement record traceable.

Make it easier

Compare the largest reading with the other three.

Take it further

Calculate the mean of the first three: about 14.07 cm, then explain appropriate rounding.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

15. Umbrellas and rainy days

A little hint

Think about which event makes people choose an umbrella.

Answer / what to notice

No. Rain commonly prompts people to open umbrellas; association alone does not establish the proposed cause.

Two things varying together can have a different causal direction or another explanation.

Make it easier

Tell the event sequence on one rainy walk.

Take it further

Invent another pair that occur together without one causing the other.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

16. Light or leaning pot?

A little hint

The pot's position supplies an alternative explanation.

Answer / what to notice

Keep pots upright and other conditions similar; compare light directions and record growth over time.

The original scene mixes light direction with starting orientation; one image cannot separate their effects.

Make it easier

Draw an upright pot as the improved starting condition.

Take it further

Explain why several similar seedlings would strengthen the comparison.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

17. Estimating a whole patch

A little hint

Find flowers per square metre, then scale up.

Answer / what to notice

Mean 3 per square metre gives about 60 daisies; it is an estimate.

Samples represent only part of the patch, so uneven distribution and sampling variation affect accuracy.

Make it easier

Add the four counts and divide by four.

Take it further

Explain why choosing only flower-rich corners would bias the estimate.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

18. Rusting versus reshaping

A little hint

Changing shape alone is different from reacting with oxygen.

Answer / what to notice

Rust formation.

Iron reacts with oxygen in the presence of water to form rust; bending mainly changes the object's shape.

Make it easier

Label the bent object same material, new shape.

Take it further

Explain why colour change can be a clue but does not alone prove every case is a chemical reaction.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

19. Compressing the air model

A little hint

Keep the same number and size of particle dots.

Answer / what to notice

It mainly reduces the spaces between particles.

A gas is compressible because its particles are relatively far apart; the simple model does not shrink the particles.

Make it easier

Draw the same ten dots in a large box and a smaller box.

Take it further

State one limitation: drawn dots are far larger than real molecules and not to scale.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

20. Equal volumes, different masses

A little hint

Density compares mass for the same volume.

Answer / what to notice

B: 1.2 grams per millilitre, compared with A's 1.0.

Divide mass by volume: $12 \div 10$ and $10 \div 10$. More mass in equal volume means greater density.

Make it easier

Compare the masses first because the volumes match.

Take it further

Predict the mass of 25 millilitres of B under the same conditions: 30 grams.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.